Sample Mark Scheme: P000350

NCFE Functional Skills Qualification in Mathematics at Level 1 (501/2325/7)

Activity 1		Marks
1A1	**Cambridge**	**2**
	Peterborough identified as 20%	1
	Cambridge identified (award both marks for Cambridge as final answer)	1
1A2	**1/3**	**3**
	Cambridge, London, Oxford, York OR 4 (places) seen OR 25% seen	1
	4/12 or their value/12 seen (accept evidence of their/12 simplified, for example, 1/4 from their 3/12)	1
	CAO (correct answer only) 1/3	1
1B1	**2:3**	**2**
	10:15 (accept 10 and 15 seen)	1
	CAO 2:3	1
1B2	**20 (%)**	**3**

Sample

Activity 1		Marks
	7/35 OR 1/5 OR 0.2 (accept 35/7 (= 5))	1
	7/35*100 OR 0.2*100 OR 1/5*100 OR equivalent	1
	CAO 20 (%)	1
1B3	**0.4 or .40 or 0.40**	**2**
	40 (%) seen OR 100 - 35 - 25 OR 100 - 60	1
	CAO 0.4 (condone display as 0.40 but do not accept 0.4 %, 0.4 percent)	1
	Total marks:	**12**

Activity 2		Marks
2A1	**31.56 (km)**	**1**
	CAO 31.56 (km)	1
2A2	**1.03 (seconds)**	**2**
	32.71 - 31.68 seen OR 32.71 and 31.68 specifically identified	1
	CAO 1.03 (seconds)	1
2B1	**30 (minutes)**	**2**
	150 (minutes) seen OR 2.5/5 (=0.5 hours)	1

Sample

Activity 2		Marks
	30 (minutes) accept 0.5 hour or 1/2 hour but appropriate units required	1
2B2	**14 (minutes)**	**3**
	720/60 or 12 seen (calories per minute)	1
	168/12 OR 168 / their calories per minute (award both method marks for 168/720*60)	1
	CAO 14 (minutes)	1
2B3	**33.33, 33.3, 33 (%)**	**2**
	1/3 (in the gym)	1
	CAO 33.33, 33.3, 33 (%)	1
2C1	**(£) 505**	**2**
	300 + 40 + 15 + (2*50) + (2*25) or equivalent	1
	CAO (£) 505	1
2C2	**(£) 215 with valid check**	**4**
	720 OR 30*12*2 OR 30*24	1
	720 - 505 (accept subtraction using their values)	1
	CAO (£) 215	1
	Check using reverse calculation, for example, 215 + 505 = 720 **FT (follow through)**	1

Sample

Activity 2		Marks
	Total marks:	**16**

Activity 3		Marks
3A	**15.2 (km/hour) with valid check**	**4**
	15 (minutes) = 0.25 (hours) OR 4 x 15 = 1 hour/60 mins OR 3.8/15	1
	3.8/0.25 OR 3.8/15*60 OR 4 x 3.8	1
	CAO 15.2 (km/hour)	1
	Check using reverse calculation, for example, 15.2*0.25 = 3.8 or 15.2/4 = 3.8 FT	1
3B1	**123**	**2**
	615/5 or their value/5	1
	CAO 123	1
3B2	**53 (may be seen in workings box)**	**3**
	172 OR 43*4 seen	1
	172 - 119 OR 43*4 - 119 OR 43*4 - 38 - 42 - 39	1
	CAO 53 (may be seen in workings box)	1
3C	**36 m^2 or 360 000 cm^2**	**3**

Sample

Activity 3		Marks
	240 (cm) converted to 2.4 (m) OR 15 (m) converted to 1500 (cm)	1
	2.4*15 seen OR 240*1500 (consistent units and correct values required)	1
	CAO 36 m^2 OR 360 000 cm^2 (correct units required)	1
	Total marks:	**12**

Overall marks:	**40**
Pass mark:	**27**

Summary of Skills Standards and Coverage and Range

(Note: where task reference and marks are indicated against a skill standard they can be for any of the associated coverage and range statements)

Skills standards	Total marks	Required weighting	Actual weighting	Coverage and range (can be covered across all skills standards)	Task reference	Marks awarded
Representing **R1** understand practical problems in familiar and unfamiliar contexts and situations, some of which are non-routine **R2** identify and obtain necessary information to tackle the problem **R3** select mathematics in an organised way to find solutions	**14**	**30 - 40 %**	**35%**	**a.** understand and use whole numbers and understand negative numbers in practical contexts	**1B3, 2C1, 2C2, 2C2, 2C2**	**5**
				b. add, subtract, multiply and divide whole numbers using a range of strategies		
				c. understand and use equivalencies between common fractions, decimals and percentages	**1A2, 1A2, 1B2, 1B2, 2A2, 2B3, 2B3**	**7**
				d. add and subtract decimals up to two decimal places		
				e. solve simple problems using ratio where one number is a multiple of the other	**1B1, 3A, 3A, 3A**	**4**
				f. use simple formulae expressed in words for one- or two-step operations		

Sample

Analysing **A1** apply mathematics in an organised way to find solutions to straightforward practical problems for different purposes **A2** use appropriate checking procedures at each stage	**13**	**30 - 40%**	**32.5%**	**g.** solve problems requiring calculation with common measures, including money, time, length, weight, capacity and temperature	**2B1, 2B1, 2B2, 2B2, 2B2, 2C1, 2C2, 3A**	**8**
				h. convert units of measure in the same system	**2A1, 3C, 3C, 3C**	**4**
				i. work out areas and perimeters in practical situations		
				j. construct geometric diagrams, models and shapes		
Interpreting **I1** interpret and communicate solutions to practical problems, drawing simple conclusions and giving explanations	**13**	**30 - 40%**	**32.5%**	**k.** extract and interpret information from tables, diagrams, charts and graphs	**1A1, 1A1, 1A2, 1B1, 1B2**	**5**
				l. collect and record discrete data and organise and represent information in different ways		
				m. find mean and range	**2A2, 3B1, 3B1, 3B2, 3B2, 3B2**	**6**
				n. use data to assess the likelihood of an outcome	**1B3**	**1**
Total marks:	**40**					**40**

Sample